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ז׳ חֶשְׁוָן תשפ״ז

What Would Settle the Leakage Attribution in Magic-State Cultivation on Willow

A reading of the grafted-code memory experiment, with a pre-registered test

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Abstract

Google's 2026 magic-state cultivation experiment on Willow grafts a cultivated magic state from a distance-3 color code into a distance-5 code that mixes color-code and surface-code stabilizers. In the supplement's memory experiment on that grafted code, the logical error rate is about seven times a depolarizing simulation that omits leakage. The authors call leakage the likely main cause, show two leakage signatures in the detector data, and ran no removal experiment. This note makes three contributions. First, a reading of the attribution: the authors trace leakage to readout parameters not optimized for a redefined measurement layout, so a readout fix that closes the gap would not by itself refute leakage. The note separates that route from two others that would, if they also explained the signatures: readout errors that never populate the |2> state, and variability in physical qubit performance, which the main text names beside leakage on an earlier grafting figure. Neither predicts the signatures on its own. Second, a record check: the public data deposit holds no files for the nine-cycle run behind the attribution, so no outside party can test it numerically. Third, a pre-registered experiment: leakage removal crossed with readout re-optimization, three-state readout in every arm, the noise strength frozen from a separate control, and pass margins fixed against a same-run baseline, with a reading assigned to each outcome. A list of figures easy to quote out of their conditions closes the note. Until that experiment runs, the attribution stays likely and not ablated.

Layman TL;DR

Google's 2026 experiment held a carefully prepared quantum state for a few rounds of error correction, and it failed about seven times more often than a simulation predicted. The team thinks the cause is leakage, a qubit escaping to a third energy level, and shows two traces of it, but did not run the fix that would prove it. Uneven quality across qubits is still a possible cause. This note explains which experiment would settle the question, what each outcome would mean, and why outsiders cannot check it from the published data.

I. The claim and what it rests on

Rosenfeld and a large Google team implemented magic-state cultivation on Willow [1], the protocol proposed by Gidney, Shutty, and Jones [2]. A noisy magic state is injected into a distance-3 color code, refined by a fault-tolerant measurement of the logical Hadamard operator with postselection, and the surviving state is then grafted into a larger encoding (distance 5, 54 qubits). The paper calls the target the surface code. During the cycles reported here the grafted code still contains stabilizers like the color code's as well as the surface code's, so it is not matchable; the main text says d - 1 cycles would be needed before switching to a matchable code, and decodes the grafting data with Tesseract, a most-likely-error decoder based on A* search [3].

Notation: d is code distance; N is the number of error-correction cycles in the grafted code; p is the noise strength of SI1000, a superconducting-inspired circuit-level depolarizing model [4] used throughout the paper's simulations, sometimes with added coherent noise; LER is logical error rate, as the supplement uses it.

Supplement claims here are from arXiv:2512.13908v1 (Dec 15, 2025), and section and figure numbers are v1's. On Oct. 18, 2026 the scored items were compared against the published Science supplement: numbers and wording are unchanged, and only the numbering moved (Sec. XIV is Sec. VIII there; Figs. S22, S23, S24, and S14 are Figs. S20, S21, S22, and S11).

The claim sits in the v1 supplement, Section XIV [1], a grafted-code memory experiment the authors call a start on studying the cost of error-correction cycles. The cultivated |T> state is idled for up to N = 9 cycles, and its LER is compared with state-vector simulations in qevol, the supplement's restricted-rank trajectory extension of the Pauli+ simulator of [5], using SI1000 at p = 2.3e-3. The experimental LER is about seven times the simulated LER. The authors state that population leakage is likely the main source of the disagreement, and give two signatures, both absent from the simulations, which do not include leakage:

- Detection-event likelihood rises slowly over time in the experimental data (Fig. S23 [1]), as long-lived population outside the computational subspace would produce.
- Detector covariance shows strong time correlations that propagate across the qubit grid (Fig. S24 [1]), which the authors describe as the characteristic leakage pattern.

The circuits contained no leakage removal units of the kind described in Google's 2023 surface-code paper [5] (the supplement's reference 49). The 2024 Willow paper [6] used a related step, data qubit leakage removal (DQLR) [7]; this note does not treat the two names as the same mechanism. The missing removal units are a precondition for the attribution, not evidence for it: they let leakage accumulate, and they would let the other causes in Section II act just as well. The non-grafted circuits agree with SI1000 at the same p: the main text reports Fig. 2 in approximate agreement and Fig. 3 in agreement with simulation, and Fig. S21 shows qualitative agreement for the kickback-tomography circuits [1]. The gap is specific to the grafted memory run. The attribution rests on the two signatures.

"Likely" is the authors' word. No run adds leakage removal units with everything else held fixed, and no measurement of |2> population on the grafted circuit has been reported.

II. Three routes to the gap

The supplement's account of how leakage arises is specific. The grafting circuit redefines which sub-lattice of qubits is measured mid-circuit; the readout control parameters were not optimized for that configuration; that led to possible frequency collisions involving qubit |2> states, and subsequent leakage [1]. Readout and leakage are therefore one causal chain in the authors' account, and listing unoptimized readout as a rival explanation misreads it. Three routes need to be kept apart:

(a) Readout control parameters, not optimized after the redefinition, cause |2> collisions, which cause leakage, which causes the gap. This is the route Section XIV names. It is the leakage claim.

(b) The same unoptimized readout causes errors that never populate |2>, such as assignment errors, and those cause the gap. Section XIV does not name this route. It is this note's alternative, and it would refute the leakage attribution.

(c) Variability in physical qubit performance. The main text names it, beside leakage, for the experiment-simulation difference in the Fig. 4 injection-only tomography [1]. Section XIV, where the seven-fold gap is reported, does not name it.

A fourth candidate sits outside the physics. Section XIV does not say the simulated curve was decoded with Tesseract, and the simulations are state-vector runs, so a decoder or model mismatch remains open.

The routes differ in what they predict for the two signatures, not only for the size of the gap. Table 1 sets them side by side.

Table 1. What each route predicts. "Gap" is the excess of experimental over simulated LER; "S23 rise" is a slow increase in detection likelihood over the circuit; "S24 spread" is time-correlated detector covariance that spreads across the grid.

Route	Gap	S23 rise	S24 spread
(a) readout, then |2> collisions, then leakage	yes	yes: leaked population is long-lived and accumulates	yes: leakage is time-correlated and can move between qubits
(b) readout errors without |2>	yes	no: assignment errors are independent from cycle to cycle	no: independent errors give no time correlation
(c) qubit variability	yes	no: a static error rate does not grow over the circuit	spatial structure only, not time correlation
Decoder or model mismatch	possibly	no: S23 is measured before decoding	no: S24 is measured before decoding

Routes (b), (c), and mismatch can each account for some or all of the gap's size, but each would need a second mechanism to account for the signatures. The signatures therefore already favor route (a) over the others. They do not prove it: other time-correlated processes, such as slow drift during the circuit, are not excluded, and the signatures were read from published figures, not from data. Routes (a) and (b) both predict that re-optimizing readout narrows the gap. Only a direct measurement of |2> population tells them apart. That is the design constraint Section IV is built around.

Leakage in topological codes has a literature of its own, from Suchara, Cross, and Gambetta [8] through McEwen and others [9] and Battistel, Varbanov, and Terhal [10].

III. What the public record can and cannot check

The Zenodo record cited in the v1 data-availability line, 10.5281/zenodo.17872003, publishes measurements.csv and detection_events.csv for Figs. 2, 3, and 4, with Fig. 4 at N = 2 and N = 3. Opened on Oct. 18, 2026, it has no directory for the N = 9 run behind Figs. S22, S23, and S24. The S23 rise and the S24 covariance can therefore be checked only by eye against the published figures. Neither can be tested numerically from the record, and no outside party can fit

a leakage model to the data that carry the attribution. Releasing the N = 9 detection events would change that at no experimental cost.

An outside simulation check is still possible, with limits. The v1 supplement cites Marshall and Kafri's incoherent approximation of leakage [11] (its references 38 and 43) for its leakage modeling; the cultivation team can apply its own implementation directly. No leakage rates have been measured on the grafted circuit, so an outside rerun would have to borrow rates from the 2023 paper [5]. That makes it an order-of-magnitude check on whether leakage at 2023-like rates can produce a gap of about seven, not a reproduction.

The 2023 calibration target is the average detection probability in [5]. After the Fig. 2 averages there, that average shows a relative rise of 12% at distance 5 and 8% at distance 3 over 25 cycles, with a typical risetime of roughly 5 cycles, and the Pauli+ simulation raises it by 7% over 25 cycles at distance 5. Those are rises in averages over stabilizers of the same weight. Fig. S23 of [1] reports a slow increase for some detectors, not an array average, so a match to the 2023 averages would not certify the shape of those traces. The 2023 target is also a different processor generation and a plain memory circuit. The grafted-circuit rise is a prediction of the check, not a fit to Fig. S23.

IV. The settling experiment

Only the cultivation team can run this. What a note can do is fix, before the run, what each outcome would mean.

Three-state readout, which resolves |2> and has been used on Google hardware before [7], [9], can run on the grafted circuit on the existing Willow processor with the team's runtime. It measures |2> population on measure qubits every cycle and on data qubits at the final readout, instead of inferring it from detector signatures. No other measurement separates route (a) from route (b), and it is cheap enough to run first, on its own.

The full test is a 2 by 2. Run the grafted-code memory experiment in four arms: leakage removal units added or not, crossed with readout control parameters re-optimized for the redefined layout or left as they were. Every arm carries three-state readout, so each arm reports its LER ratio, its cycle-resolved |2> population on measure qubits, and its final |2> population on data qubits.

Four design rules hold for every arm:

1. Fixed p. Freeze p from an unpostselected memory circuit on the same qubits on the same day, and use that one value for every arm. Do not fit p to the grafted data; a per-arm fit could shrink the ratio by construction. The postselected kickback-tomography circuit of Fig. 3 serves as a cross-check, not as the calibration, because its 8% retention makes a fitted p depend on the postselection rule. Because p is re-frozen, the baseline arm B need not reproduce the v1 ratio of about 7; every reading below is taken against B in the same run.
2. Matched simulation. The simulation for each removal arm includes the removal-unit operations at the same SI1000 strength, so that the overhead those units add sits in the denominator as well as the numerator.
3. Pass margin. The run counts only if B shows a gap: the lower 95% bound on B's LER ratio must exceed 3. Fix the shot count in advance so that the 95% interval on each arm's ratio is no wider than plus or minus 15% of B's observed ratio; if B's interval comes out wider, add shots before reading any other arm. SI1000 is itself simplified, so a ratio of exactly one is not the target. Call an arm closed when the upper 95% bound on its ratio is below 2; unchanged when its interval overlaps B's; widened when its lower bound exceeds B's upper bound; narrowed otherwise.
4. |2> criterion. Call an arm leakage-reduced when its mean |2> population on measure qubits over cycles 2 to 9, and its final |2> population on data qubits, are each below half of B's, with non-overlapping 95% intervals.

Table 2. Pre-registered readings of the 2 by 2. Arms: B, baseline (no removal, readout as in v1); R, readout re-optimized only; L, leakage removal only; RL, both. "Closed", "narrowed", "unchanged", and "widened" follow rule 3; "|2> down" follows rule 4. A pattern not listed is reported as observed and scored inconclusive.

Pattern	Reading
L and RL closed; R not closed	Leakage confirmed by removal; its entry is not the readout path Section XIV names.
R closed with |2> down; L closed	Route (a) confirmed: readout misconfiguration caused leakage, which caused the gap. Leakage confirmed, with its entry path named.
R closed without |2> down	Route (b): the readout path causes the gap without leakage. The leakage attribution fails.
L or RL narrowed, not closed	Leakage is one cause among several. The residual goes to route (b), route (c), or decoder or model mismatch.
All arms unchanged	The attribution fails. The signatures in Figs. S23 and S24 [1] need another explanation.
L or RL widened	Removal overhead dominates at this scale despite rule 2. Inconclusive, and reported as such.

The authors say they expect improvement from leakage removal, better calibration, and larger distances. Follow-up work can be read against Table 2 by which row it reports, and by whether it reports the rerun at all.

V. Numbers to cite with care

Several figures from the cultivation paper are easy to quote out of their conditions.

- The headline factor. The Science abstract, retrieved on Oct. 18, 2026, says cultivation reduces the error by a factor of 40, with a state fidelity of at least 0.9999(1), retaining 8% of attempts. The v1 preprint gives injection-only error 4.4(2)e-3, which the main text calls "similar to our physical qubit gate errors" [1], and cultivated full-circuit error between 1e-4 and 2e-4: a ratio of about 22 to 44. "About 40" sits at the low-error end. The v1 abstract states the fidelity as 0.9999(1), without "at least".
- The seven-fold gap. Fig. S22 labels the experimental panel (b) "Exp. fit, LER = 5.3%" and the simulation panel (e) "Exp. fit, LER = 0.8%"; "Exp. fit" reads as an exponential fit. The caption gives LER "as a function of N" and does not say whether these are per cycle. The 0.8% carries one significant figure, so the ratio lies between about 6.2 and 7.1 from rounding alone. The gap is one figure, one grafted configuration, and at most nine cycles. The factor of 40 and the factor of seven are different measurements: state error against injection, and memory LER against SI1000.

Figure 1 draws that gap beside the two rules in Section IV.

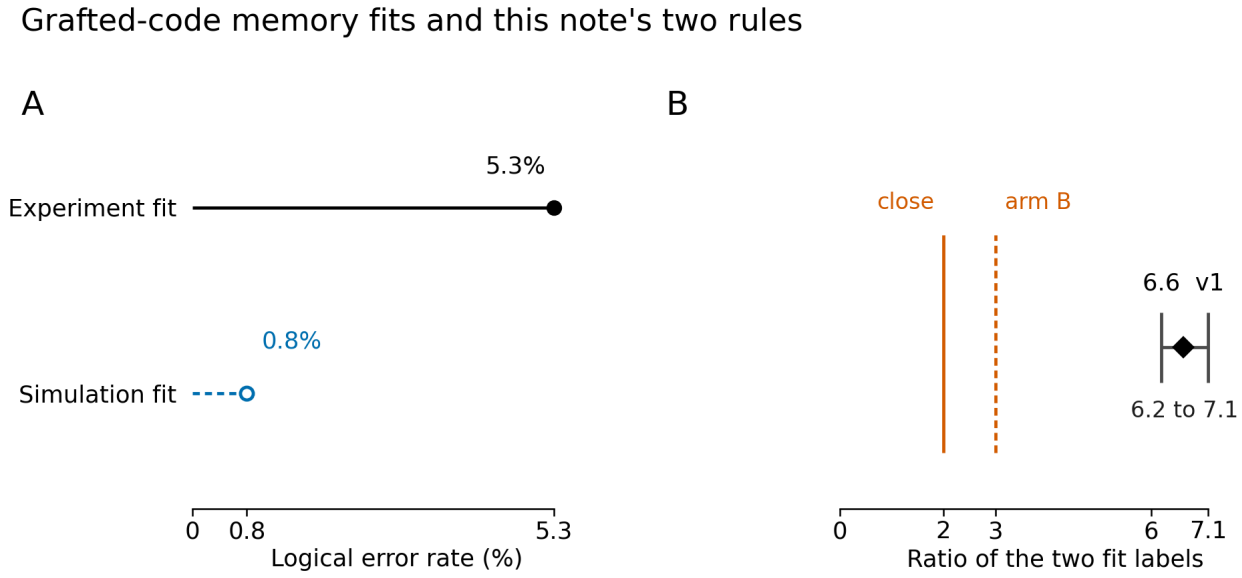


Figure 1. Left: the Fig. S22 exponential-fit labels, experiment 5.3% and SI1000 simulation 0.8%, drawn from zero. That caption does not say the percents are per cycle. Right: the ratio 5.3/0.8 = 6.6. The capped bar is only the rounding span of the one-significant-figure 0.8% (6.2 to 7.1). It is the published v1 ratio. The rerun is read against same-run baseline B, which need not equal 6.6. The solid line at 2 is the close line: an arm is closed when its upper 95% bound is below 2. The dashed line at 3 is the gate: the run counts only if B's lower 95% bound exceeds 3. Both lines were fixed before any rerun. No 2 by 2 arm has been run. A ratio of 1 is not the target. This is not the factor-of-about-40 state-error comparison.

- Two noise strengths. The main text uses p = 3.0e-3 for Fig. 2 and p = 2.3e-3 for Figs. 3 and 4, attributes the change to shifted qubit frequencies and different control calibrations, and reports logical-error variation of about 1e-4 over an hour. Section XIV carries p = 2.3e-3 over from Fig. 4; the seven-fold gap is relative to that value.
- The residual detuning bound. Early tomography of cultivated states returned non-physical norms, which the supplement attributes to low-frequency phase noise and which led the team to build kickback tomography. With echoing, a simulation bounds the remaining |T> infidelity from quasi-static detunings near 1e-5. That bound assumes plus or minus 200 kHz quasi-static detunings and no depolarizing noise, and Fig. S14c still shows non-physical norms for some echoed draws.
- Distance to application targets. The paper's introduction states that infidelities of about 1e-10 are needed to factor an RSA-2048 key, citing Gidney [12], and about 1e-6 to 1e-8 for some simulation and optimization tasks. The measured value, near 1e-4, is about six orders of magnitude from 1e-10. The supplement shows cultivated error scaling as p cubed in simulation. My own extrapolation from that fit, at fixed distance 3: halving p would give about 1.3e-5 to 2.5e-5, still above the 1e-6 to 1e-8 band. Neither [1] nor [2] makes that projection, and the protocol [2] reaches its low error rates at larger distance.

VI. Context: the 2024 inventory

This note grew out of the author's 2024 note on Willow [13], which listed six error channels that the simplified Pauli model behind Google's 2024 predictions [6] omitted or treated thinly. Read against that list, the cultivation experiment bears on one channel, leakage, and only through the attribution examined above. It does not test crosstalk, which appears only as a candidate source of detunings; its correlated detector features are tied by the authors to leakage and so add no independent evidence; it reuses the gate and readout baseline; and it does not address classical chaos [14]. The third-party objections tracked in [13] (noise sensitivity [15], [16]; nonlinear errors

[17]; reachability deficits [18]; classical chaos [14]) are not tested, and their 2024 grades, read from the author's mirror of that note [19], stand unchanged. None of the 2024 channels was unknown to Google, whose 2024 error budget already carried leakage.

VII. Limitations

The author wrote the 2024 note in Section VI. All figure pins and the "likely" wording are from arXiv:2512.13908v1; the published Science supplement was compared on those items only. The Zenodo record holds no N = 9 data, so the signatures were read from the published figures, not recomputed. I did not rerun the paper's simulations. The repository behind this note holds a distance-3 calibration run of a local leakage simulator; it is not a result of this note. The pass margin and |2> criterion in Section IV are my choices, fixed here before any rerun. The attribution examined here rests on one grafted-code memory figure, one configuration, and at most nine cycles.

VIII. Conclusion

The seven-fold gap in the grafted-code memory experiment has a named cause, leakage, that the authors call likely, support with two signatures, and have not ablated. Their own account puts readout misconfiguration upstream of the leakage, so the decisive question is whether the gap travels with |2> population. Route (b), qubit variability, and decoder or model mismatch remain open. Two steps would settle it: release the N = 9 detection events, and run the 2 by 2 with three-state readout under the rules fixed in Section IV.

Availability

This note's PDF is the Zenodo paper record. The version of record is https://doi.org/10.5281/zenodo.23455419. The concept DOI, which resolves to the latest PDF, is https://doi.org/10.5281/zenodo.23452622. An OSF mirror of this PDF is https://osf.io/qme2w/ (10.17605/OSF.IO/QME2W). Zenodo is the version of record. The local simulator, the Figure 1 script, and the review files of this note are a separate software record: version https://doi.org/10.5281/zenodo.23452625, concept https://doi.org/10.5281/zenodo.23452624. The repository is https://github.com/chokmah-me/willow-follow-up-note. Third-party source pins and journal peer-review files stay local claim-checking copies. They are not in either Zenodo record, and this note's CC BY 4.0 license does not cover them. A distance-3 calibration of the simulator is not a result of this note.

AI Utilization Statement

This note was prepared with AI assistance, directed by the author at every stage. Muse (Meta) agents (model version not recorded) located and read the primary sources, cross-checked numbers and attributions in the draft against them, and drafted the text at the author's instruction. Grok 4.6 applied the author's 2023 rise-leg substitution to the v3 Section V wording, then a later critical-review pass (abstract citations, table captions, v3 Section V hardware split, Limitations local-instrument line). The leakage simulator in the repository was written by OpenAI GPT-6 Astra through OpenRouter from the author's briefs, then reviewed and tested before use; a distance-3 calibration run of it is recorded in the repository and is not a result of this note. The draft then went through two internal adversarial passes (a spec-faithful claims trace and an independent adversarial review), and two external critical evaluations commissioned by the author through OpenRouter, from Moonshot Kimi K3 and Anthropic Claude Opus 5.5. Claude (Anthropic), through Claude Code, ran a further critical review of v3 and applied its findings to produce v4, checking each change against the local copy of arXiv:2512.13908v1, and at the author's direction restructured v4 into v5 around the attribution test, then reviewed v5 and applied its findings to produce v6. Every finding from those passes was adjudicated before any revision was applied. The review files are in the software record https://doi.org/10.5281/zenodo.23452625. The author selected all sources, made every publish and revise decision, and takes responsibility for the claims and the conclusions. AI output was treated as draft throughout. Claims here were checked against

arXiv:2512.13908v1 and the Science abstract [1], and the scored items were compared against the published Science supplement. Grok 4.7 drew Figure 1 and the whiteboard sketchnote from numbers already in this note. Neither is a new measurement. No AI system is listed as a co-author.

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